

FLAT DESIGNS: PAST AND PRESENT

Be it Microsoft's Metro UI or iOS introspecting on its interface, or even tech giants like Facebook, Google, Lenovo changing their logos to make them more flat. What is behind flat design. Here's everything you should know. <http://dgit.in/FlatDesigns>

GPS, THAT HAS BEEN ALWAYS INACCURATE

GIS researcher Peter Rancher with his team explains why GPS has always been overestimating distance and has been adding extra distance to increase its reliability and still managed to go unnoticed every time! We kid you not! <http://dgit.in/GPSError>

MICROBLOGGING TO GO SHORTER?

A picture is worth a thousand words and adding a caption to it is all that is required for your next 'blog' post on image sharing social media websites like Instagram. How are these minimalistic image sharing apps getting into blogging? <http://dgit.in/InstaBlogg>

THE YOUTUBE 'BOT-KINGS'

Read here, the story of how YouTube view generating bots are still alive and how a seven second non-sensible video got thousands of views in no time, despite the video sharing's legendary stand on non-organic views and bots. <http://dgit.in/YTVBot>

SHORTS

Absentia VR unveils Tesseract 3D HMD at IGX

The device promises conversion of any sort of digital content into 3D, without lag or need for major processing power

– By Jayesh Shinde

At the recently concluded Indian Games Expo (#IGX) in Mumbai, we came upon a stall with no name or number. Quite strange. Its occupants resembled young college kids, all dressed in black T-shirts with "Absentia VR" inscribed on it. Of course, we couldn't see this tiny detail easily, nor their booth. Like a swarm of fireflies attracted to a beacon in the dark, a flood of people had congregated inside the cramped space, each one eager to try out Absentia VR's relatively unknown invention – the Tesseract. Everyone wanted to see "the plane beyond."

A company started by a gang of dedicated BITS Goa dropouts, Absentia VR has created Tesseract – a head-mounted display



expect any VR HMD to be. Just like the Oculus Rift, Sony's PlayStation VR, HTC Vive, Gear VR, and other HMD projects that are currently in the works (and slated for release in 2016), the Tesseract looks like a futuristic eyewear straight out of a sci-fi movie. However, its inventors believe it has several unique distinguishing factors that make it more lucrative (for investors) and palatable (for the general audience).

Decoding the Tesseract

The Tesseract HMD is made of plastic with cushioned lining (where it comes in contact with the head) and adjustable straps to mount the unit. It has two dials on its side to individually adjust the focal length of the lens inside (near and far) and on top it comes with an interpupillary adjustable dial to move the two lenses left or right. It has a range of sensors including and not limited to the gyro and the accelerometer. The entire system is powered by USB and an unknown ARM Cortex chip makes it tick.

The Tesseract – currently supporting 1080p and 1440p visuals – made several claims. Shubham Mishra, Co-Founder, Absentia VR, and the captain of this fledgling hardware startup, talked about some of the advantages the Tesseract has over its competitors. Unlike the Oculus Rift and other high-end VR headsets, the Tesseract doesn't need specialised content – it can upscale pre-existing 2D content to 3D without any lag. Their patent-pending algorithm differs with Oculus' implementation, too, claims Shubham and his team – where the Oculus uses predictive head tracking to pre-render pixels, the Tesseract uses smart vectorisation to define the edges and depth of an object in real-time. This approach,



for an interactive, immersive, up close and personal experience with moving visuals unlike anything you've ever experienced. Unless you've tried the latest pre-release build of the Oculus Rift (which we have).

From what took about a year from conception to first bootstrapped prototype, Absentia VR's Tesseract is what you would



according to Absentia VR, eliminates a hint of drift suffered by Oculus Rift.

Another achievement Shubham and his young team proudly proclaim is the fact that the Tesseract does no extra graphics computation on its part, relying completely on the PC's discrete GPU and resources for all the visual information fed through